# CI/CD Pipeline for Autonomous Energy Researcher Agent (EnergyMind)

Course Name: DevOps

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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*Academic Year: 2025-2026*

## Problem Statement

The energy sector generates a massive amount of information every day from multiple sources such as research publications, government policies, market trends, renewable energy reports, and real-time news updates. This information is highly fragmented and continuously evolving, making it difficult for analysts, researchers, and decision-makers to gather accurate insights quickly. Traditional manual research methods require significant time and effort to collect, filter, analyze, and summarize relevant data. As a result, decision-making is often delayed, and important trends or risks may be overlooked.

Additionally, existing search engines provide raw information but lack contextual analysis and structured reporting. Users must manually evaluate the credibility of sources, extract meaningful insights, and prepare reports. This process is inefficient, error-prone, and not scalable, especially when handling large volumes of data.

Another challenge lies in maintaining and deploying AI-based applications designed to automate such research. Manual deployment processes often lead to inconsistencies across environments, downtime during updates, and difficulty in scaling the system. Without proper containerization and continuous deployment mechanisms, managing updates and ensuring system reliability becomes complex.

Therefore, there is a need for an intelligent, automated system that can:

- Automatically filter relevant energy-related queries
- Collect data from reliable sources
- Analyze information using AI agents
- Generate structured research reports
- Maintain accuracy through validation mechanisms
- Deploy continuously with minimal downtime
- Scale efficiently using containerization and cloud infrastructure

To address these challenges, this project proposes **EnergyMind AI Agents**, an autonomous multi-agent research platform integrated with a CI/CD pipeline. The system automates the entire research workflow, from data collection to report generation, while leveraging Docker, Kubernetes, and AWS for scalable deployment. This solution reduces manual effort, improves research accuracy, and ensures continuous delivery of updated features with high availability.

## Project Objectives

The primary objective of this project is to design and develop an autonomous AI-driven research platform for the energy sector and implement a robust CI/CD pipeline for continuous integration and deployment. The system aims to automate the process of collecting, analyzing, and generating structured research reports while ensuring scalability, reliability, and security.

**1. Deploy Using Kubernetes (K3s)**

To orchestrate Docker containers using Kubernetes (K3s) for better scalability, fault tolerance, and zero-downtime deployment. The system should automatically restart failed containers and manage load efficiently.

**2. Improve Performance Using Caching**

To implement caching mechanisms that store previously generated research results in the database. This reduces API calls, improves response time, and optimizes system performance.

**3. Enable Continuous Monitoring and Updates**

To design the system in a way that allows easy monitoring, logging, and updates without interrupting user access, ensuring high availability and maintainability.

These objectives collectively aim to build a fully automated, scalable, and secure AI-based research system integrated with modern DevOps practices.

## Scope of the Project

The scope of the EnergyMind AI Agents project is to design and develop an AI-powered research platform that automates energy sector analysis and report generation. The system allows users to submit energy-related queries and receive structured research reports generated through a multi-agent workflow. The project includes the development of a Streamlit-based frontend for user interaction and a FastAPI backend for handling API requests and coordinating AI agents.

The scope also covers integration with external APIs for data collection and natural language processing, enabling automated extraction of trends, risks, and insights. A SQLite database is used to store user information, chat history, and cached research results to improve performance. The project further includes implementing secure authentication using JWT and proper error handling mechanisms.

Additionally, the application is containerized using Docker to ensure consistent deployment across environments. A CI/CD pipeline is implemented using GitHub Actions to automate building, testing, and deployment of the application. The system is deployed on AWS EC2 using Kubernetes (K3s) for container orchestration, scalability, and fault tolerance.

Overall, the project scope includes AI-based research automation, full-stack development, database management, containerization, CI/CD pipeline implementation, cloud deployment, and security integration.

## Proposed Solution

To address the challenges of manual energy sector research and complex deployment processes, this project proposes **EnergyMind AI Agents**, an AI-powered autonomous research platform integrated with a CI/CD pipeline. The system automates the complete workflow of collecting, analyzing, and generating structured research reports for energy-related queries.

The proposed solution uses a multi-agent architecture where specialized AI agents perform different tasks. The Gatekeeper agent validates whether the query is related to the energy domain. The Researcher agent retrieves relevant data from external sources using search APIs. The Analyst agent processes the collected information and extracts meaningful insights such as trends, risks, and opportunities. The Writer agent generates a structured report, and the Reviewer agent verifies the report quality before delivering the final output to the user.

The system includes a Streamlit-based frontend for user interaction and a FastAPI backend to handle requests and coordinate the AI workflow. A SQLite database is used to store user details, chat history, and cached reports for improved performance. JWT-based authentication ensures secure access to the application.

To ensure consistent deployment, the entire application is containerized using Docker. A CI/CD pipeline is implemented using GitHub Actions to automatically build Docker images, push them to Docker Hub, and deploy them to AWS EC2. Kubernetes (K3s) is used for container orchestration, enabling scalability, fault tolerance, and zero-downtime deployment.

This proposed solution automates research tasks, reduces manual effort, improves accuracy, and ensures continuous deployment of the system with high availability and scalability.

## Key Features

### 1. Multi-Agent AI Workflow

- Uses specialized agents for research, analysis, and report generation
- Modular architecture for easy updates and maintenance
- Improves accuracy through reviewer feedback loop

### 2. Automated Research Generation

- Accepts energy-related queries from users
- Fetches data from multiple sources automatically
- Generates structured and readable reports

### 3. CI/CD Pipeline Automation

- GitHub Actions builds and deploys automatically
- Ensures continuous integration and delivery
- Reduces manual deployment effort

### 4. Docker Containerization

- Application packaged into Docker containers
- Ensures consistent environment across systems
- Simplifies deployment and scaling

### 5. Kubernetes Deployment

- Uses K3s for container orchestration
- Provides auto-restart and fault tolerance
- Enables zero-downtime deployment

### 6. Secure Authentication

- JWT-based user authentication
- Passwords stored in hashed format
- Protected API endpoints

### 7. Database Management

- SQLite used for storing user data
- Chat history maintained for users
- Caching improves performance

**8. User-Friendly Interface**

- Streamlit dashboard for interaction
- Easy query submission
- Displays generated research reports

**9. Performance Optimization**

- Caches previously generated reports
- Reduces repeated API calls
- Improves response time 🚀

## Overall Architecture / Workflow

EnergyMind AI follows a **client–server architecture integrated with DevOps pipeline**, where responsibilities are divided between the user interface, backend services, AI orchestration layer, database, and cloud deployment infrastructure.

The complete flow is:

**User → Streamlit Frontend → FastAPI Backend → LangGraph Workflow → Database → Docker Container → Kubernetes (K3s) → AWS EC2 → Response**

Additionally, CI/CD automation ensures continuous deployment:

**Developer → GitHub → GitHub Actions → Docker Hub → AWS EC2 (Kubernetes) → Updated Application**

This architecture provides scalability, automation, fault tolerance, and zero-downtime deployment.

**Detailed Workflow**

**1. User Query Submission**

- User enters energy-related query in Streamlit UI
- Request sent to backend for processing

**2. API Communication**

- Streamlit sends POST request to FastAPI backend
- JWT authentication validates user

**3. Cache Verification**

- Backend checks SQLite database for cached results
- If available, response returned directly

**4. Workflow Triggering**

- If no cached data found, LangGraph multi-agent workflow starts
- Query processing begins

**5. Gatekeeper Validation**

- Gatekeeper agent verifies energy-domain query
- Invalid queries rejected

**6. Live Research Phase**

- Research agent fetches data using Tavily API
- External sources are queried in real time

**7. Analytical Processing**

- Analyst agent extracts insights
- Identifies trends, risks, and opportunities

**8. Report Generation**

- Writer agent generates structured report
- Output formatted for readability

**9. Quality Review Loop**

- Reviewer agent checks report quality
- If FAIL → returned to writer
- If PASS → workflow continues

**10. Follow-up Suggestions**

- Suggestion agent generates related research queries
- Improves user exploration

**11. Data Persistence**

- Final report saved in SQLite database
- Chat history stored
- Cached for performance

**12. Response Delivery**

- Structured response returned to frontend
- Displayed to user

## DevOps CI/CD Workflow

**13. Code Commit**

- Developer pushes code to GitHub repository
- Changes tracked using version control

**14. Continuous Integration (CI)**

- GitHub Actions automatically triggered
- Application build process starts
- Docker image created

**15. Docker Image Push**

- Docker image pushed to Docker Hub
- Latest version stored in registry

**16. Continuous Deployment (CD)**

- AWS EC2 server pulls latest Docker image
- Kubernetes (K3s) updates running containers

**17. Kubernetes Orchestration**

- K3s manages pods and services
- Failed containers automatically restarted
- Rolling update ensures zero downtime

**18. Application Available to Users**

- Updated application deployed on AWS EC2
- Users access latest version automatically 🚀

## Tools & Technologies Used

**Core Development Stack -**

- **Python** – Primary programming language used for backend logic, workflow implementation, and AI integration.
- **FastAPI** – High-performance REST API framework used to build backend services and handle HTTP requests efficiently.
- **Streamlit** – Frontend framework used to create an interactive and user-friendly research interface.

**Database -**

- SQLite
- SQLAlchemy ORM

**DevOps Tools -**

- Docker
- Kubernetes (K3s)
- GitHub Actions
- AWS EC2

**Security -**

- JWT Authentication
- CORS Middleware
- Environment Variables

## Screenshots

## Reports / Dashboards / Models

EnergyMind AI is designed to generate structured, decision-ready research outputs rather than simple text responses. The system produces executive-level reports enriched with analytical insights, quality validation, and persistent storage for future reference. The outputs are suitable for academic, industry, and strategic research use cases.

## Reports & Outputs Generated

- **Structured Executive-Level Research Reports**
  Professionally formatted reports including summary, analysis, insights, and conclusions.
- **Industry-Focused Insights**
  Extracted trends, risks, opportunities, and market observations based on real-time data.
- **Automated Review & Refinement**
  Reports are evaluated by a reviewer node to ensure clarity, logical structure, and completeness before final delivery.
- **Database-Stored Research Records**
  Final outputs are stored in the KnowledgeBase database for caching and future retrieval.
- **Text-Based Backup Files**
  Reports are also saved as .txt files inside the knowledge_base folder for backup and traceability.

## Model Used

- Llama-3.1-8B-Instant **via** Groq **API**
  The system uses the Llama-3.1-8B-Instant large language model for fast and efficient inference.
  Groq provides optimized high-speed LLM execution, enabling quick report generation and smooth multi-agent workflow processing.

## Key Outcomes

- **Automated Research Generation**
  The system successfully generates structured energy research reports automatically, reducing manual effort and saving time.
- **Implementation of Multi-Agent Architecture**
  Specialized AI agents work collaboratively to perform query validation, data collection, analysis, and report generation.
- **CI/CD Pipeline Integration**
  Automated build and deployment achieved using GitHub Actions, Docker, and Kubernetes for continuous delivery.
- **Scalable Cloud Deployment**
  Application successfully deployed on AWS EC2 using Kubernetes (K3s) ensuring scalability and fault tolerance.
- **Secure Authentication System**
  JWT-based authentication implemented to ensure secure user access and data protection.
- **Performance Optimization using Caching**
  Database caching reduces repeated API calls and improves response time.
- **Containerized Application Deployment**
  Docker used to package application ensuring consistent environment across development and production.
- **User-Friendly Research Interface**
  Streamlit dashboard provides easy query submission and structured report visualization.

## Conclusion

The **EnergyMind AI Agents** project successfully demonstrates the development of an autonomous research platform integrated with modern DevOps practices. The system automates energy sector research using a multi-agent workflow that collects, analyzes, and generates structured reports. By integrating FastAPI for backend services and Streamlit for frontend interaction, the platform provides a user-friendly and efficient research environment.

The project also implements containerization using Docker and automated CI/CD pipeline using GitHub Actions, ensuring continuous integration and deployment. Deployment on AWS

EC2 with Kubernetes (K3s) provides scalability, fault tolerance, and zero-downtime updates. Additionally, secure authentication, database caching, and error handling improve system performance and reliability.

Overall, the project achieves its objective of reducing manual research efforts while ensuring automated deployment and scalability. It demonstrates practical implementation of AI-based automation combined with DevOps tools, making the system efficient, secure, and production-ready.

## Future Scope & Enhancements

- **Support for Multiple Domains**
  The system can be extended to handle research queries beyond the energy sector such as healthcare, finance, and technology.
- **Advanced Database Integration**
  SQLite can be replaced with scalable databases like PostgreSQL or MongoDB for handling large volumes of data.
- **Improved User Interface**
  The Streamlit frontend can be enhanced or replaced with a React-based UI for better user experience and responsiveness.
- **Real-Time Data Streaming**
  Integration with live data sources and streaming APIs to provide real-time analytics and insights.
- **Multi-Node Kubernetes Deployment**
  Upgrade from single-node K3s to multi-node Kubernetes cluster for higher scalability and load balancing.
- **Monitoring and Logging Integration**
  Add monitoring tools such as Prometheus and Grafana for performance tracking and system health monitoring.
- **Role-Based Access Control**
  Implement different user roles such as admin, analyst, and viewer for better security and management.
- **Report Export Functionality**
  Allow users to download generated reports in PDF or Excel formats.
- **Auto Scaling Implementation**
  Configure Kubernetes auto-scaling based on traffic and workload.

- **CI/CD Pipeline Enhancements**

  Add automated testing and security scanning in CI/CD pipeline for improved reliability.